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1. Executive Summary

The purpose of this device is to give aquaponic, hydroponic, home gardeners, and anyone interested in growing their own nutrients a tool to automate their systems, which is then modifiable via webpage. Aquaponics is a growing method, using aquatic livestock (such as freshwater shrimp and fish) in a livestock tank that circulates water (more advanced systems employ solid matter filters) through/on a grow medium submerging roots allowing the plants to uptake aqueous nitrogen as nutrients. This device supplies 8 electric sockets able to independently operate on user defined schedules and modifiers (cutting electric to devices during low water events). Users set when devices operate via webpage that will then set when the outlets supply electricity to their devices. This technique increases the efficiency of water used (10% of that of traditional agricultural methods) which in turn could interest countries/areas where water is in short supply, as well as military bases which could use these systems to supply troops with better quality nutrition.

1.1 Purpose of this document

The purpose of this document is to describe in sufficient detail how the Automated Aquaponic Garden PLC system is to be constructed. The System Design Document translates the Requirement Specifications into a document from which the developers can create the actual system. It finds the top-level system architecture, and finds hardware, software, communication, and interface components.

1.2 Design Scope

We are developing and building a dynamic indoor automated aquaponic garden plc that uses a web-based interface to automate the system components, both on the fly and using pre-installed operation designs that were perfected for different flora and fauna living/growing in this system. We would like to eventually use farming and aquaponic databases to design system patterns that would allow users with only a basal knowledge of gardening to find success with these systems. (I.E., the pump would run for longer periods for plants that thrive with more water (such as lettuce and chard) and adjustable based on what type of plants the end user was growing). Our solution allows 8 different and independently controlled electrical outlets using a micro-computer / PLC that controls both the outlets and serve interactive web pages that verified & authorized users would use to direct and control the electricity to the aquaponic electrical components. A sample garden uses pipes that happen to get flooded by independent pumps, an aerator, lighting, and automated feeding. The web page allows users to run schedules on equipment, store & query individual data (growth, yields, schedule logs, etc.), analyze stored data to suggest best scheduling, and optionally import open-source agricultural databases when newer a version is uploaded. This allows users to get the most optimized schedules by running the algorithms on larger datasets (improving precision as more units gathered data and optimization algorithms were tweaked) to improve the optimization.

1.3 Intended Audience and Document Overview

Our intended audience are home gardeners, anyone with a limited water supply, as well as anyone interested in using scheduled devices. If you are a client/user we would recommend section 1.4, and sections 3.6-3.10. If you are a project sponsor, we would recommend 3.1-3.5 and all of section 4.

1.4 Definitions, Acronymns, and Abbreviations

Appliance- Any external device connected to this product by way of electrical cord for the purpose of being powered by this product.

Appliance Type- a user-defined specification of the purpose of an attached device. Types consist of, but are not limited to, lights, aerators, and water pumps.

Browser- External software intended for supplying standard access to user-readable data from a server.

Current Web Browser- Up-to-date releases of Safari (v. 14.0) and Chrome (v. 89.0 [Android] and 87.0 [iOS]) mobile web browsers as well as Chrome (v. 89.0), Edge (v. 89.0), Firefox (v. 86.0), and Safari (v. 14.0) desktop web browsers.

Persistent Memory- Memory stored on a non-volatile micro-SD card, recoverable in the case of power loss.

Tank- The aquarium or water-based system used in conjunction with the device. This is where the livestock is kept.

Growth Data – The rate of growth of livestock/produce.

Yield Data – The final weight of harvested livestock/produce.

Grow Schedule – The schedule of the multiple components of a given grow bed based on the plant types.

Accurate (Time) – Time will be considered “accurate” if the system time is within 300 seconds of UTC time adjusted for the time zone offset.

ISO – International Organization for Standardization

IEC – International Electrotechnical Commission

IEEE – Institute of Electrical and Electronics Engineers

NEMA – National Electrical Manufacturers Association

C/LM – LAN/MAN Standards Committee

LAN/MAN – Local Area Network and Metropolitan Area Network

1.5 Document Conventions

Table colors are defined by this mapping:

Green part able to perform task

Orange part able to perform task with modifiers

Red part only performs task with deep modifiers

1.6 References and Acknowledgments

<https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/>

https://www.bbc.com/future/ bespoke/ follow-the-food/ why-our-food-needs-to-use-less-water/?utm_source=pocket-newtab

<https://www.health.harvard.edu/blog/backyard-gardening-grow-your-own-food-improve-your-health-201206294984>

<http://www.fao.org/in-action/globefish/market-reports/resource-detail/en/c/1263851/>

2.3 Alternative Approaches

There are many alternative approaches that would cost arguably less, such as using a solar powered electric pump that would pump water to a collection area that would gravity water the plants as the bucket filled (this method would only operate during the day and in areas where sunshine is present such as Baja California).

2.4 Impact of Success

Imagine the effects of empowering others to grow their own nutrients and spending less water in agriculture seem like that would have a net positive effect, with no blatant negative impacts other than transitioning grocery sales towards building these systems. Another positive effect is detailed by the above BBC source.

3 Context of Design Solution

3.1 Design Objectives

Objective Prime: The designed invention becomes as integrated into the modern-day kitchen as an oven, refrigerator, and sink. The designed invention is a leader in automating the aquaponic industry.

The hobby of indoor gardening is growing, especially in North America where the industry is noticing a burst of spending on indoor gardening during this pandemic. This device allows for people to automate and allow for variance of component operations in their aquaponic systems, using a webpage interface to manage scheduling various operations.

Objective 1: Device allows users to independently operate 8 different American style electrical sockets using a webpage app.

This is one of the core objectives of this device switches what operations are running by supplying the electrical current to the aquaponic equipment such as the pumps, aerator, and lights. Scheduling is applied to the operations using the webpage-based interface. This device would enable multiple variants of growing environments. This is allowing user to tweak their grow beds settings to imitate their plants' best-known environments. Data on the socket usage (such as the type of component (e.g., pump, aerator, heater), operating schedule, timed devices, other POIs) is generated by user input that is then uploaded to the system to adjust the optimization scheduling. The number of grow beds and system type (ebb and flow, floating, etc.) is specified by the user via data input. The webform would then dynamically change the data inputs for the sockets when users changed their grow beds / aquaponic systems.

Objective 2: This device senses when the livestock tank water level is low and stops operations until water is added to the required level.

This would then send the primary user a notification to their email on file that this system's water level was low and required filling.

Objective 3: This device would host a webpage that allowed users to operate electric sockets on this device, run automated scheduling for lights, aeration, and water pumps, store growth data based on user input, log system modifications (i.e., when sockets were operational, water level notifications sent), and recommends best known grow schedules based on plant types set by the user.

This is another core objective is this enables communication between a user and the device's scheduling. Implementing an area to store growth data should encourage engagement of users and their aquaponic systems. Giving the system data about

vegetative states such as seedlings, vegetation, and flowering, would allow the system to adjust lighting based on each grow beds' plant types.

Objective 4: This device would communicate with a server database that would send push notices that new versions of the database and optimal schedules were available to download.

Giving users new datasets to add vegetables/fruits/herbs to their garden should entice users to download new versions, and improved optimization schedules would increase the chances of larger yields.

Objective 5: This device would include a prototype aquaponic system. This aquaponic system is getting built as a component of this device to show functionality. The design uses a pipe system that would allow for up to 107 plants and would measure, allowing for up to 2cm of variance, the following:

Size: 100x50x110cm/38"x19"x43" (of each pipe)

Hole Size: 32mm

Tube Diameter: 63mm

Width: 19 in

Length: 43 in



Source:

https://www.gardensupplie.com/index.php?main_page=product_info&products_id=1681&msclkid=b7292ced432e17f62aec4cc825426253

This is the aquaponic system we intend on reverse engineering, then modifying with multiple aquaponic components. This specific design details will get more descriptive as this invention matures, to demonstrate every function/ability of this pairing of systems. This prototype's design will define the size of tanks, pumps, other essential components that are omitted on this document as details are described in the design document.

3.2 Design Assumptions

Assumption 1: The designed appliance will be powered by a GFCI compliant outlet.

Assumption 2: The designed appliance will be directly used by devices with three-prong connections.

Assumption 3: The designed appliance will be used with 120-volt (+/- 5%) devices.

Assumption 4: The designed appliance will be used indoors.

Assumption 5: The designed appliance will not be used above 50% humidity.

Assumption 6: The designed appliance will not be submerged in water.

Assumption 7: The designed appliance will be used at temperatures above 40 degrees Fahrenheit.

Assumption 8: The designed appliance will be used at temperatures below 100 degrees Fahrenheit.

Assumption 9: The designed appliance will be connected to an IEEE 802.11 wireless network.

Assumption 10: The designed appliance will be secured to a surface with screws no shorter than 1 inch.

Assumption 11: The designed appliance will be used in a clear space of at least 1 foot by 2 feet.

Assumption 12: The designed appliance will be used in conjunction with an aquarium that allows filling and draining.

Assumption 13: The designed appliance will be used in compliance with supplied instructions.

3.3 Design Requirements

Objective 1: Device allows users to independently operate 8 different American style electrical sockets using a web app.

Objective 2: This device senses when the livestock tank water level is low and stops operations until water is added to the required level.

Objective 3: This device would host a webpage that allowed users to operate electric sockets on this device, run automated scheduling for lights, aeration, and water pumps set by the user.

Objective 4: This device would communicate with a database that would store growth data based on user input, log system modifications (i.e. when sockets were operational, water level notifications sent), and advises grow schedules based on plant types set by the user.

Objective 5: This device would include a prototype aquaponic system. This aquaponic system is getting built as a component of this device to demonstrate functionality. The design uses a pipe system that would allow for up to 107 plants and would measure, allowing for up to 2" of variance, the following:

Grow bed structure:

Size: 48" * 72" * 36" (up to 2" of variation on this)

Req No.	Obj No.	Requirement
10	1,2,3,4	Device shall modify current flow independently on 8 electrical sockets via webpage interface. This operation is superseded by safety mechanisms to prevent unsafe operations, i.e. safety features could interrupt signals until safety measurements are met.
11	1,3	The device shall host a webpage that provides controls to the onboard outlets. This website shall get structured by the specifics of this instructable: https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/ that will then get designed and modified for this device's uses.
12	1,3	The webpage shall allow internal assignment of appliance types to individual outlets.
20	1,2,3	Users can set a minimum water level that shuts down water pump operations during low levels as a safety measure. A minimum water level is defined by the user by setting a sensor (e.g. https://www.adafruit.com/product/464) Onto the side of their livestock tank submerging the top measurement point to their system's best running water level and could modify their warning and shutoff levels using settings on their account.
21	2	If the water does go below the minimum water level, a liquid tape measure sensor will sense it and shut down water pump operations.
22	2,3	Users will get sent a notice when water levels are within range, that users are able to modify, of set minimum water level. (via email)
30	3	The webpage shall be compatible with current mobile web browsers.
31	3	The webpage shall be compatible with current desktop web browsers.
32	3	The webpage shall meet coding standards of PHP.

33	3	The webpage shall supply a dropdown of appliance types for each outlet.
34	3	The webpage shall support persistent memory of assigned appliance types.
35	3	The webpage shall allow the user to add and remove custom appliance types to the available list.
36	3	The webpage shall allow the user to optionally set minimum and maximum tank water levels.
37	1,3	The webpage shall allow for recurring daily scheduling of individual outlets.
38	1,3	The webpage shall allow for scheduling of outlets based on appliance type.
39	1,3	The webpage shall provide a menu for adding and modifying outlet schedules.
40	1,4	The database shall be hosted on the raspberry Pi.
41	3,4	The database shall store growth and yield data.
42	3,4	The database shall store system logs.
43	3,4	The database shall store grow schedules.
44	3,4	The system shall allow users to input growth data.
45	1,3,4	The webpage shall allow for scheduling to be done with the accuracy of whole minutes.
46	3,4	The device shall maintain a right system time.
		The system shall use a network time server to maintain accurate system time.
		System time shall be right as long as a network connection is maintained, and the network server is right.
47	3,4	The webpage, using login data on the database, shall provide unique access to the owner of the device.
48	1,3	The device shall supply instructions for the user on getting and changing webpage access, using, and maintaining the prototype aquaponic system, and other instructions.
50	5	Prototype system shall consist of multiple pipes grow beds with the measurements shown above. Variants of aquaponic system types such as ebb & flow and floating shall get demonstrated on this prototype system. Variance is allowable on up to 2" as long as system is watertight, other than water loss from evaporation, plant usage, and other "normal" aquaponic operations.
51	1,5	Prototype system shall use multiple water-pumps, aeration devices, and lights to demonstrate the operation capabilities of the automated aquaponic plc device. For this prototype each grow bed shall get an individual water pump and set of lights. These components shall operate using the schedule/settings defined on the plc and set by the user.
52	5	Prototype system shall operate as a growth medium / environment for various fruit bearing and other types of vegetation, as well as varied species of fish/shrimp depending on nitrogen load dependencies.
53	5	Prototype system shall promote environments that are optimal for the species kept in the livestock water tank. Parameters for the livestock and plants such as temperature, nitrogen and o2 levels, and other environmental elements shall get met by the system components and inputs (i.e., water, food, medicine).
54	5	Prototype system shall use livestock tank water to flow through the various grow beds to give the flora life nutrients.

55	5	Prototype system shall get built using aquaponic practices and grow bed types.
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3.4 Design Constraints

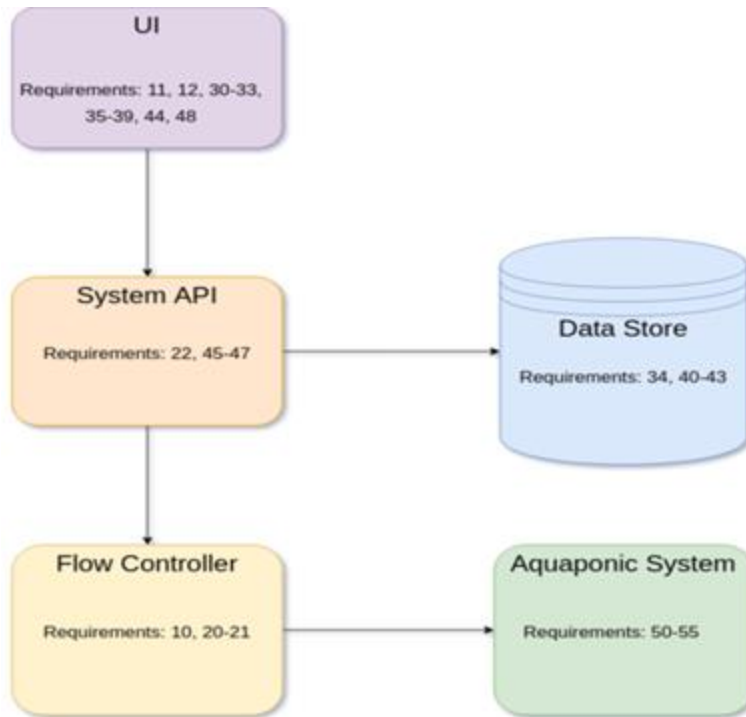
Const No.	Constraint
10	The electric system device shall get built using a modified version of the following design using the following components: • Raspberry Pi with SD Card • Micro USB Power supply (Make sure your power supply can supply a full 1 amp) • 8-Channel 5V Relay board - EBay (About \$10 shipped here.) • 4 (qty) 15-Amp Power Receptacles (Lowe's or Home-Depot, get the cheap ones about 80 cents each) • 18AWG Solid Hookup Wire (3 colors) - Menards (About \$18.50) • Short length of 14-Gauge wire. (Only need about 15 inches of white, I found some scrap) • Large wire-nuts • Colored jumper wires - Female to Female buy them on EBay here. • cord • Scrap wood pieces • wood screws • Optional (but handy): A female end of an extension cord or broken holiday-light set. These parts are used to build a small web enabled electric strip cord.
11	The wiring for this and the gpio on the raspberry pi zero wh is found on the following website: https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/ . The website gets described later in this document showing the modifiers that apply to this device, such as the gpio scripts are written in perl. The raspberry pi behaves as it's own server for this component.
20	The aquaponic prototype shall comprise of a livestock tank of ~ 50 U.S. gallons of a roughneck Tupperware storage tote. Such as https://www.homedepot.com/p/HDX-55-Gal-Tough-Storage-Bin-in-Black-HDX55GONLINE-4/205597365
30	The Automated Aquaponic Garden PLC system will have a maximum budget of \$200.
40	The Automated Aquaponic Garden PLC system will have a maximum power consumption of 20 amp.

3.5 Design Standards

Stand. No	Standard
10	ISO/IEC 12207:2008 – Systems and software engineering – Software life cycle processes, ISO, 2008
20	ISO/IEC/IEEE 26511:2018 – Systems and software engineering – Requirements for managers of information for users of systems, software, and service, ISO/IEC JTC 1/SC 7, 2018
30	NEMA 5/IEC 60906-2:2011 – IEC system of plugs and socket-outlets for household and similar purposes, PEL/23, 2011
40	IEEE 802.11ac-2013 – IEEE Standard for Information technology – Telecommunications and information exchange between systems, C/LM, 2013

3.6 Design Functionality

1. User Interface
 - a. Allows user interactions with the system. Interactions include logging data, setting schedules, monitoring plant growth, configuring associative devices.
 - b. Submits user data to the System API to process.
2. System API
 - a. Supplies an intermediary between the user interface, data store, and flow controller.
 - b. Process data from the User Interface
 - c. Stores data in the Data Store
 - d. Actuates the Flow Controller
3. Flow Controller
 - a. Receives signal from system API and allows or denies current flow.
 - b. Alters the power being sent to different devices within the Aquaponic System.
 - c. Signals are sent via GPIO of the Raspberry PI to their mapped channel (socket ids (user socket names e.g., “grow bed 2 aqua pump”) link to individual channels on an 8-channel relay board, which then open/closes current flow to the socket on the electric strip board.
 - d. Water level sensor is user installed on the livestock tank. Users then set their desired level to send emails when low water levels are sensed. The raspberry pi once given a signal from this water level sensor, then can shut down operations (specified by the user via UI) and send emails notifying users of low water levels).
4. Data Store
 - a. Stores data necessary to keep system functioning properly, such as log and scheduling data.
5. Aquaponic System
 - a. Supplies an environment for the growth of aquatic livestock, vegetation, and/or flowers for agricultural purposes.



3.7 Design Impact

1. Cultural

Automated aquaponic systems being used in underdeveloped countries could affect the economy enough to see the country adopt new cultures or reevaluate current cultural norms; these changes may be positive or negative.

2. Economic

Positive: Farms could reduce the cost to operate which could lead to more profit. This larger profit margin could reduce the amount of farm subsidies granted which would affect the national budget. Fewer imports of livestock as these could get farmed in a domestic market.

3. Environmental

Positive impacts would include lower water usage for agriculture, lower pesticide use, lower fertilizer runoff into public water tables, increase agricultural yield through layering, allows agriculture in barren areas.

Negative impacts would include trash from bad units; mitigations would include refurbishing units that went bad, stripping materials from bad units, and using safe e-waste methods.

4. Global

Positive impacts would include the above environmental impacts, especially in areas which water is less abundant. This current day publishing defines why this is important : https://www.bbc.com/future/ bespoke/ follow-the-food/ why-our-food-needs-to-use-less-water/?utm_source=pocket-newtab

Other positive benefits include those bulleted in this webpage:

<http://www.fao.org/3/V5290E/v5290e02.htm>

Such as:

- *Enough nutritious food, including some staple foods, for all the family year round.* This also includes extra food stocks for processing and sale to obtain income, and a reserve for special occasions or emergencies (e.g., sometimes a staple food crop is lost in a flood, eaten by pests, or reduced because the farmer falls sick and cannot work for a period).
- *Income from the sale of home garden produce.* Sale from home garden produce can make a substantial contribution to a family's income (to buy daily essentials and farming inputs that cannot be produced on the family's farmland as well as other goods and services).
- *Important farm development activities take place in the home garden.* Some farm inputs come from home garden activities, for example plant propagation, raising and housing animals, and making or repairing tools. The home garden is also a place for trying out new crops and farming techniques.
- *The home garden is the most direct means of supplying families with most of the non-staple foods they need year round*

Negative impacts would include those in the agricultural industry, as this would lower demand for purchased food.

5. Public health

Positive Impact: With aquaponic systems, you can guarantee there is no possibility the food can contract e-coli. Fish are cold blooded animals and cannot host E-coli.

Encourages better personal health discussed in the following :

<https://www.health.harvard.edu/blog/backyard-gardening-grow-your-own-food-improve-your-health-201206294984>

Potential Mitigations for Negative Impact: More data and research.

6. Public safety

Positive Impact: You do not have to use any toxic chemical or fertilizers or pesticides.

Negative Impact: One concern is growth of the plants is the effect of insect pests.

Pesticides are not a

practical answer in dealing with problems with insects in aquaponic environments.

7. Public welfare

Positive Impact: With a moderate initial cost and a low upkeep cost, a household aquaponic system can reduce the cost, and increase the supply, of fresh produce to the individual.

Negative Impact: Potentially higher water and electricity utility costs per household.

Potential Mitigations: Efficient scheduling suggestions can help reduce the power draw of the system.

8. Social

Positive Impact: Inexpensive aquaponic systems can provide a more accessible and consistent source of healthy produce for the general public.

Negative Impact: Widely available aquaponic systems could reduce the demand for local and/or organic produce, adversely affecting orchards and farms.

Potential Mitigations: Farms could focus more heavily on produce that cannot be generated through aquaponics.

3.8 User Characteristics

The most important class of users will be the ones who are wanting to grow crops for sustenance or livelihood. These users will have frequent interaction with the product and will be the users most reliant on post-purchase technical support. These same users are likely familiar with the crops and/or livestock they will be raising,

The secondary user would be hobbyists intending to learn about aquaponics. They may not be as involved as the class above, but they still expect to be able to achieve moderate success with limited experience. This user would not want to spend every day maintaining a system for which they do not expect to see unknown results, and so would rely heavily on the automation of the product.

The third class would be the users that purchase the product but do not intend to use it for an aquaponics system. These users will likely have specific projects they want to automate and will use socket assignment of the webpage in a more flexible way. These users may still require moderate support in setup and troubleshooting. Because additional products are still a valid use of the product, it is important that the accessibility for users extends beyond single-use.

3.9 Operating Environment

The environment for which our product will operate will be indoors. The exact location of where indoors will be up to the user but the space will need to be big enough to fit the device itself along with a livestock tank. The temperature will need to be appropriate for the survival of livestock and crops, but will function best between 65°F and 75°F. The device itself does not create a significant amount of mechanical stress, but the location of its use needs to support the weight of the livestock tank when filled with water.

3.10 User Documentation

This product will be delivered with a paper instruction manual included. The instruction manual will be a packet, written in English, covering the following:

- Explanation of aquaponic process, goals, and benefits.
- Safety warnings and safety standards.
- Parts list of all provided components.
- Parts list of suggested components for building the non-electrical system.
- Instructions on both building the provided system and its suggested use.
- Instructions on connecting the processor to a wireless network.

- Instructions on access the built-in webpage and creating a user account.
- Instructions on function and use of the webpage, including troubleshooting, type assignment, scheduling, and settings.
- Examples of settings and their applications within the system.
- Limited product guarantee.
- Contact information for user support.

4 Technical Approach

To achieve the requirements and constraints outlined in section 3, several hardware and software decisions had to be made. For the computer, a Raspberry Pi Model 3 A+ was chosen due to its wireless capabilities and processing speed. For the water sensors, the Adafruit and Seed sensors were chosen for their accuracy and ability to easy connect to the circuit board of the computer. The Elegoo relay board is the most cost-efficient and applicable component of the options for connecting the computer to the electrical sockets. The electrical sockets from Legrand fit the criteria best, based on the ability to modify the connection between individual sockets, and a significantly lower price.

The approach to software focused less on cost and more on the functionality of different options. SQLite was chosen for the database because it does not require a server to function, it handles storing and retrieving blob data, and the license is in the public domain. For writing the operational code, which must be able to access the computer, the SQL database, and the webpage, PHP was chosen. For the front-end, user interface portion of the software, JavaScript was chosen to create a webpage in the minimal Nevada style.

4.1 Hardware

1. Circuit Board/Single-Board Computer
Design Choice: Raspberry Pi Model 3 A+

	Design Options			
Criteria: Circuit Board/Single-Board Computer:	Raspberry Pi Model 3 A+	Raspberry Pi Zero WH (Zero W with Headers)	Raspberry Pi Zero W	Raspberry Pi 4 Model B - 1 GB RAM
Req.10: Must support installation of an Operating System.	Raspberry Pi OS	Raspberry Pi OS	Raspberry Pi OS	Raspberry Pi OS
Req. 40: Must support expandable storage/memory (i.e., USB or SD).	True	True	True	True
Const. 10: Must support a Micro USB power supply rated for 1A.	True	True	True	True
Const. 30: Cost	25.00	14.00	10.00	35.00
Req. 10: Must support integrating Relay Board (GPIO)	True	True	Must solder GPIO	True

To allow for new feature that enable/use 5 GHZ wireless and increased speed, this design is using Raspberry Pi Model 3 A+, with Raspberry Pi Zero WH (Zero W with Headers) as the runner up due to price.

2. Water Sensor: Temperature

	Design Options			
Criteria	SEN-11050 Waterproof Sensor - Sparkfun	DS18B20 Digital Temperature Sensor - Adafruit	QAP1030.200 Temperature Sensor - Siemens	101990019 Temperature Sensor – Seed
Const. 30: Cost	11.95	9.95	9.56	7.50
Const. 11: Must support integrating with Circuit Board/Single- Board Computer GPIO	True	True	True	False
Req. 53: Must be able to detect temperatures from 15°F - 150°F	-67°F to 185°F True	-67°F to 257°F True	-13°F to 203°F True	-67°F to 257°F True
Req. 53: Accuracy	+/- 0.9°F	+/- 0.9°F	+/- 2.5°F	+/- 0.9°F
Req. 20: Waterproof	True	False	False	True

Design Choice: DS18B20 Digital Temperature Sensor from Adafruit.

The DS18B20 is the most fitting temperature sensor overall. While it is not the least expensive, it does meet the requirements for integration with our Circuit Board and temperature range as specified in the Design Trades (Worksheet 4.01).

3. Water Sensor: Water Level

	Design Options			
Criteria	Optomax Digital Liquid Level Switch	ES10010 Elecall Float Switch	Grove Water Level Sensor - Seeed	eTape Liquid Level Sensor - Adafruit
Const. 30: Cost	17.47	6.98	6.95	39.95
Const. 11: Must support integrating with Circuit Board/Single-Board Computer GPIO	True	True	True	True
Req. 20: Must detect maximum and minimum water levels	False – Can detect acceptable and too high OR too low, not both	True	True	True
Req. 20: Must be able to set acceptable levels using interface	False – Requires physical location change to detect	False – Requires physical modification	True	True

Design Choice: Grove Water Level Sensor from Seeed

The Grove Water Level Sensor provides a cheap, accurate method of detecting the physical water level at any time. Unlike the float switch, which only senses 3 points (acceptable, high, low), it transmits constant data of the level relative to the sensor.

4. Relay Board

	Design Options			
Criteria	OONO 4 SPDT 10Amp Power Relay Module	Electronics-Salon 8 Channel 10Amp SPDT Power Relay Module Board	ELEGOO 8 Channel DC 5V Relay Module with Optocoupler	Huayao 2pcs 4 Channel DC 5V Relay Module with Optocoupler
Req. 10: Must support 8 channels.	False	True	True	True
Req. 10: Must support integrating with Circuit Board/Single-Board Computer GPIO	True	True	True	True
Const. 30: Cost	9.90	38.00	10.99	10.88
Const. 10: Must be rated for 5V	True	True	True	True
Req.11	N/A	True	True	False

Design Choice: ELEGOO 8 Channel DC 5V Relay Module with Optocoupler this most align with this design.

5. Power Receptacles

	Design Options			
Criteria	Legrand radiant® 15 Amp Black Self Test Tamper Resistant GFCI Outlet 1597TRBKCC4	Legrand® TradeMaster® 15-Amp Light Almond Duplex Outlet 3232LACC30	Smart Electrician® 15-Amp Tamper Resistant Light Almond Dual USB Charging Duplex Outlet M9253LA	Smart Electrician® 15-Amp Tamper Resistant White Dual USB Charging Duplex Outlet - 2 Pack TR-BAS15-2USB
Const. 40: Must be rated for 15A	True	True	True	True
Const. 30: Cost	13.34	.43	9.34	17.79
Const. 11: Must support disconnection (brass tabs)	False	True	True	False

Design Choice: Legrand® TradeMaster® 15-Amp Light Almond Duplex Outlet 3232LACC30 due to price.

We intend on getting 1 of the 15 Amp USB outlets to direct current to the raspberry pi.

4.2 Software



1. Database

	Design Options			
Criteria	SQLite	MariaDB	LucidDB	Microsoft Access
Req. 40: Linux Compatible	Yes	Yes	Yes	No
Req. 40: Has SQL Interface	Yes	Yes	Yes	Yes
Req. 41: Max Blob Size	2 GB	4 GB	None	1 GB
Req. 44: Ease of Use (by operations available)	Easy	Easy	Easy	Moderate
Const. 30: License Type	Public Domain	GPL	GPL	Proprietary
Req. 40: Requires Server	No	Yes	Yes	Yes

Design Choice: SQLite

SQLite is the optimal choice for this project. Although MariaDB meets many of the same requirements as SQLite, SQLite does not require a server to be running to access a database (database is instead stored in a file), allowing for quicker access and less processing power required. Additionally, SQLite is open-source and highly compatible with the Raspbian operating system.

2. Backend Programming Language

	Design Options			
Criteria	PHP	Python	Perl	Bash Scripting
Req. 10: Must be able to make system calls via the Operating System	Yes	Yes	Yes	Yes
Req. 40: Must be able to make calls to the selected database	Yes	Yes	Yes	Yes
Req. 47: Must support calls from the frontend application	Yes	Yes	Not easily	Not easily
Req. 10	Satisfied	Not satisfied	Not satisfied	Not satisfied

Design Choice: PHP

PHP is the language of choice due to it being the only language that satisfied requirement 11, "...the website shall get structured by the specifics of this instructable: [https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/ ...](https://www.instructables.com/Simple-and-intuitive-web-interface-for-your-Raspbe/)"

3. Frontend Programming Language

	Design Options	
Criteria	PHP	JavaScript
Req. 10	Satisfied	Satisfied
Req. 11	Satisfied	Satisfied
Req. 12	Satisfied	Satisfied
Req. 30	Satisfied	Satisfied
Req. 31	Satisfied	Satisfied
Req. 32	Satisfied	Not satisfied
Req. 33	Satisfied	Satisfied
Req. 34	Satisfied	Satisfied
Req. 35	Satisfied	Satisfied
Req. 36	Satisfied	Satisfied
Req. 37	Satisfied	Satisfied
Req. 38	Satisfied	Satisfied
Req. 39	Satisfied	Satisfied
Req. 45	Satisfied	Satisfied
Req. 47	Satisfied	Satisfied

Design Choice: JavaScript

JavaScript does not meet requirement 32, however, due to limitations of PHP (i.e., client-side processing), JavaScript will be used and requirement 32 will be removed.

4. User Interface

	Design Options				
Criteria	Dashio : A Bootstrap Admin Template	Organic Store : An Organic Farming Website Template	Vanilla : One-Page CSS Template	Nevada: Minimal One-page HTML Template	Minimal Dark – Small and Simple HTML Template
Req. 11: Supports design of entire system	True	True	True	True	True
Req. 48: Supports easy flow to users (easy navigation)	True	True	True	True	True
Req. 11: Simplifies user interface with the system functions	Intermediate	Difficult	Intermediate	Easy	Easy

Design Choice: Nevada: Minimal One-page HTML Template

Nevada: Minimal One-page HTML Template was the best choice since the time needed to implement the user interface was simple and easy along with it being able to simplify the user interface with the system functions and supports the design of the entire system as well as easy navigation.

5 Appendix: Test and Evaluation Master Plan and Report

Requirement	Test Method	Evaluation Method	Threshold	Objective
10, 11, 12	Microprocessor, electrical outlets, wireless connection, internet-connected device, multimeter	Output of sockets will be measured against different conditions through the web interface.	120V +/-5% output through active sockets, no output through inactive sockets	120V output through active sockets, no output through inactive sockets
20, 21, 22	Microprocessor, electrical outlets, wireless connection, internet-connected device, water level sensor, water, multimeter	Output of sockets will be measured against different conditions through the water level sensor	120V +/-5% output through active sockets, no output through inactive sockets	120V output through active sockets, no output through inactive sockets
30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 45, 47	Microprocessor, wireless connection, internet-connected devices of differing OS and browsers	Web accessibility will be determined through consistent interfaces and persistent database population across different browsers	User input is stored in database, affects microprocessor operation, is restored to user 90% of the time	User input is stored in database, affects operation, and is restored to the user upon new login regardless of device or browser 99% of the time
40, 41, 42, 43, 46	Microprocessor, developer console	Data will be stored and retrieved from database	Data is accurately stored and retrievable 90% of the time	Data is accurately stored and retrievable 99% of the time
48, 50, 51, 52, 53, 54, 55	Aquaponic system components, microprocessor, electrical outlets	Operation of appliances through the microprocessor control will be visually confirmed	Lights and pumps operate appropriately 90% of the time	Lights and pumps operate appropriately 99% of the time

Constraint	Test Method	Evaluation Method	Threshold	Objective
10, 11	Microprocessor, aquaponic system components, electrical outlets, multimeter	Device will be visually inspected to ensure correct composition of parts	The microprocessor and electrical outlets are appropriately connected	All components of the system are appropriately connected
20	Livestock tank	Visual inspection of appropriate material	Open-top container of at least 45-gallon capacity	Open-top container of 50-gallon capacity
30	Total components of system	Total cost of all parts will be summed	Total cost is less than \$200	Total cost is less than \$160
40	Microprocessor, electrical outlets, system components, multimeter	The amperage of the system in use will be measured	Consumption is under 20 amp	Consumption is under 18 amp

Standard	Test Method	Evaluation Method	Threshold	Objective
10, 20, 40	Microprocessor, developer console	Visual inspection of code	All standards are met	All standards are met
30	Electrical outlets	Visual inspection	All standards are met	All standards are met

Summary Tables of Test Results

Requirement	Test Date	Test and Evaluation Location	Result	Notes	Date Passed (Accepted)
10, 11, 12					
20, 21, 22					
30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 45, 47					
40, 41, 42, 43, 46					
48, 50, 51, 52, 53, 54, 55					

Constraint	Test Date	Test and Evaluation Location	Result	Notes	Date Passed (Accepted)
10, 11					
20					
30					
40					

Standard	Test Date	Test and Evaluation Location	Result	Notes	Date Passed (Accepted)
10, 20, 40					
30					

Test and Evaluation Methods

Requirements (verification of achievement)

Requirements 10, 11, and 12 will be tested at Wright State which has an electrical connection, a wireless network, and a multimeter. The test method is as follows: The microprocessor and outlets will be powered; a separate device will be connected to the processor's webpage through a wireless connection. The electrical output of the outlets will be evaluated using the multimeter under variable conditions where the user tells the processor to switch on or off specific sockets. The threshold is within 5% of 120V and the objective is 120V output for sockets that should be active, no electrical output for sockets that should be dormant.

Requirements 20, 21, 22 will be tested at Wright State which has an electrical connection, a wireless network, and a multimeter. The test method is as follows: The microprocessor and outlets will be powered; the water sensor will be placed in an appropriate amount of water; a separate device will be connected to the processor's webpage through a wireless connection. Specific sockets will be designated as non-emergency outlets by the user through the webpage. The water level measured by the sensor will be lowered, triggering the shutoff of all non-emergency outlets and the output of all outlets will be evaluated. The threshold for active outlets is within 5% of 120V and the objective is 120V. For shut off sockets, the threshold and objective are both no current.

Requirements 30-39, 45, and 47 will be tested at Wright State which has an electrical connection, a wireless connection, and computers with Windows and Linux operating systems. The test method is as follows: The microprocessor will be powered, and devices will connect at different times to the processor through the wireless connection. A device will access the webpage of the processor and changes will be made to the input and state of the processor. A different device will then connect, and the persistence of the previous data will be evaluated as well as the availability of the webpage itself. The threshold is that 90% of the time, input data is stored and available upon a new connection. The objective is that all data is stored, measured by availability at least 99% of the time.

Requirements 40-43, and 46 will be tested at Wright State which has an electrical connection and a computer to access the microprocessor directly. The test method is as follows: The microprocessor will be powered, and the computer will have a direct connection to the processor's memory. Code from the processor will be executed to store specific data, such as growth rates and system logs, that do not affect the operation of the processor directly. This data will then be retrieved from the processor's database and evaluated. The threshold is that this data is recoverable 90% of the time while the objective is that the data is always recoverable, measured by a consistency of at least 99%.

Requirements 48, 50-55 will be tested at Wright State which has an electrical connection, and where the parts of the system will be assembled. The test method is as follows: The microprocessor and outlets will be powered, and the appliances of the system will be connected to the outlets. The operation of these appliances will be visually observed and evaluated by appropriate operation of each appliance. The threshold is that all appliances

operate normally at least 90% of the time. The object is that the appliances always operate correctly, measured by normal operation at least 99% of the time.

Constraints (verification of conformity)

Constraints 10 and 11 will be tested at Wright State which has a multimeter, and where the system will be assembled. The test method is as follows: the microprocessor and outlets will be powered, and the appliances will be connected to the active outlets. Through visual inspection and measurement with the multimeter, the correct composition of parts will be evaluated. The threshold and the objective are that all the parts are connected appropriately, based on the system diagram.

Constraint 20 will be tested at Wright State where the system will be assembled. The test method is as follows: The container to be used as the livestock tank in the prototype will be evaluated based on its volume. The threshold for this volume is 45 gallons, and the objective for the volume is 50 gallons.

Constraint 30 will be tested at Wright State where the system parts will be purchased. The test method is as follows: The sum of the cost of all parts will be evaluated before purchase. The threshold is a cost below \$200, and the objective is a total cost under \$160 to allow for replacement parts if needed.

Constraint 40 will be tested at Wright State which has an electrical connection, a multimeter, and is where the system will be assembled. The test method is as follows: The microprocessor and outlets will be powered, and the system components will be connected. The amperage of the entire system will then be evaluated with the multimeter. The threshold is for the consumption of the whole system to be under 20 amps. The objective is for the consumption of the system to be under 18 amps.

Standards (verification of compliance)

Standards 10, 20, and 40 will be tested for compliance at Wright State which has an electrical connection, and a computer to connect directly to the microprocessor. The code will be visually evaluated through the development console to ensure that standards ISO/IEC 12207:2008, ISO/IEC/IEEE 26511:2018, and IEEE 802.11ac-2013 are met by the code.

Standard 30 will be tested for compliance at Wright State where the system will be assembled. The wiring of the outlets will be visually inspected to ensure that standard NEMA 5/IEC 60906-2:2011 is met. Specialized equipment will not be required.

6 Appendix: Résumés of Team Members

ALEXANDER GENTRY

25 E. Cottage Ave. West Carrollton, OH 45449 | C: (937) 475-4041 |
gentryaj@mail.uc.edu

Education

Wright State University GPA: 3.60 August 2020 - Present

- **Major:** Bachelor of Science in Computer Science
- **Coursework:** Graphics, Database Management, Technical Communications, Operating Systems
- Expecting to graduate in Fall 2021

University of Dayton GPA: 4.00 August 2019 - May 2020

- **Major:** Bachelor of Science in Computer Science
- **Coursework:** Game Development, Discrete Structures, Computer Architecture, Statistical Math, Operating Systems

Sinclair Community College GPA: 4.00 January 2016 - May 2019

- **Major:** Associate of Science with a focus in Computer Science; High Honors
- **Coursework:** Java, C++, Linux, Calculus, Differential Equations, Linear Algebra
- Winner of the 2016-2017 AMATYC Math Competition at Sinclair

Employment

University of Dayton Research Institute January 2019 - Present
Software Technician with Sensors & Software Systems Division

- Developed a working event driven agent composed of Java adapters and Generic Modeling Environment behavior models with the purpose of communicating with a simulation and operator.
- Assisted in development of event specifications and message handling as a service to be used on current and future projects within the group.
- Created a series of Clingo tutorials within Confluence to assist new hires in learning ASP. Tutorials included step-by-step instructions for solving specific logic problems, as well as images, explanations, and example code.
- Worked in several small teams combining applications consisting of Java, Clingo, Python, GME, Groovy, and C++ in an Agile-like environment using Jira and GitLab.

DERRICK W. KALINEY(937) 853-6469 | dvwkaliney@gmail.com**EDUCATION**

Bachelor of Science: Computer Science (Currently Attending - Target Graduation December 2021)
Wright State University – Dayton, OH

AAS in Computer Information Systems: Software Development Track
Sinclair Community College – Dayton, OH

TECHNICAL SKILLS

- | | | | |
|----------------|----------------------|-------------------------|------------------------------------|
| • JavaScript | • Ubuntu | • Git | • MySQL |
| • Ajax | • GitHub | • Assembly (HSC12) | • SFTP |
| • Java | • Oracle | • C / C++ / C# | • Apache2 |
| • ASP.NET | • Agile | • PHP | • Cloud Based Server Installations |
| • Bootstrap | • Project Management | • Nginx | |
| • Unit Testing | • Perl | • Load Balancing | |
| • TDD | • SQL | • Proxy / Reverse Proxy | |
| • HTML | • AWS | • SSH | |

RELATED WORK EXPERIENCE

Wright State University - Computer Science Teaching Assistant / Student **2020 – Present**

- Provide insight to aspiring computer engineers/scientists to ensure success of lab and individual goals
- Ensure standardization for assembly/c language-based solutions on multiple individual projects
- Collaborate and contribute to multiple code bases to ensure student success
- Utilize multiple platforms for effectively communicating with various groups
- Setup and manage group and individual discussions to aid with labs and their solutions
- Lead and provide guidance for engineering better solutions using assembly and c languages.
- Weekly code reviews of numerous projects (roughly 40-70 individual labs per week)

TEKSystems – Software Developer **2017 – 2019**

- Full stack development and maintenance for legacy ASP.net applications utilizing an Oracle database
- Presented and implemented new messaging and bug tracking systems to the client team to help improve team collaboration and communication, using tools such as Slack and Trello.
- Ensured quality control through rigorous testing of all new features and patches, including refactoring
- Created design and user documentation to communicate updates to the product owners and users
- Developed and modified stored procedures for the accompanying Oracle database

MarketVision Research – Research Associate / Web Developer **2016 – 2017**

- Wrote Perl scripts to build data collection websites

University of Dayton Research Institute – Junior Programmer **2015 – 2016**

- Contracted for LexisNexis in maintenance and improvement of their legacy metadata organization program
 - Java based legacy desktop application using Swing for front-end development
 - Created SSRS reports for accompanying SQL database
 - Additionally, worked and configured an error reporting system to deploy from an Apache Tomcat Servlet providing web API services utilizing a RESTful architecture (Eagle Application)
- Developed in house, for the Department of Justice, a web project for a responsive ASP.NET website and mobile app to help organize service providers for various groups and organizations.
 - Utilized Bootstrap and CSS in order to keep website stylish and responsive in relation to accompanying native app
 - Implemented jQuery and JavaScript to perform various tasks, such as filtering presentation data and DOM modification

Chalmer Frueauf

8305 Hyannis Port Drive Apt 1B

Centerville, OH 45458

Cell: 513-850-0933

cfrueauf@rainbowdata.com

Professional Objective

Hardworking Cyber Security professional dedicated to high-quality performance and efficiency. Great communicator and team player with organized detail-oriented mindset. Highly accurate and able to learn new tasks quickly.

Education

May 2018 – Present, Wright State University

- Majored in Information Technology and Cyber Security (BS)
- Expected Graduation – Spring 2022

June 2014 - April 2016, Ohio University

- Played collegiate football for The Bobcats
- Studied Computer Science while enrolled

2014 High School Diploma, Archbishop Moeller, Cincinnati OH

- All Ohio Honors (2012 – 2014)

Training/Certificates

- Cyber Awareness Challenge
- Insider Threat Awareness
- Counterintelligence Awareness and Security Brief

Skills Highlights

• Cyber Security • Information Technology • Operating Systems • Website Configurations	• Database (MySQL) • Java • JavaScript • HTML/CSS	• Perl • ASP/VBScript • C++
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Professional Experience

February 2017 – Current: Rainbow Data Systems Incorporation, Computer Programmer

- Configured websites for different Government Contracts
- Database Configurations (MySQL)
- Develop programs in different languages to complete tasks that are required (Perl, ASP, HTML, CSS)
- Network Fundamentals